

Examiner reports

Q1.

The vast majority of candidates showed they understood the requirements of this question although often algebraic errors were made.

Many candidates could not find the derivative of $\frac{1}{2t}$ correctly in part (a), the 2 moving between the numerator and denominator position to various powers; the derivative of $\frac{1}{t}$ itself was usually correct. However, most candidates used the chain rule correctly to find a gradient. Some candidates actually found a cartesian equation before differentiating, with some completing successfully.

In part (b) most candidates went on to use their gradient to find that of the normal, and bar a few slips in calculating the x and y coordinates, were successful in finding a correct equation, using their gradient.

In part (c) most candidates found t from $x = 4t + 3$ and substituted in $y = \frac{1}{2t} - 1$, often without any simplification, which was quite acceptable. Those who did attempt some simplification often made algebraic errors, but these were condoned following a correct substitution. Those who attempted an alternative method of eliminating t were more prone to error. For some candidates, poor presentation led to $\frac{1}{2t}$ becoming $\frac{1}{2}t$.